

Lab03 assignment

For this lab, you will draw flags from around the world.

The original version of this assignment was developed by [Fawzi Emad](#) at the University of Maryland.

Getting started

1. Download the zip file from Google Classroom.
2. Unzip / extract / uncompress the zipfile.
3. Rename the folder from lab03 to username-lab03 but replace "username" with your email username. So for example, mine would be jspacco-lab03
4. Navigate into the project folder, find the file named lab03.codeworkspace, and double click it to launch the project in VS Code
5. In the tmp folder, find the .jar file that corresponds to your computer and operating system. Copy it from the tmp folder to the lib folder.
 - app-windows.jar is for Windows, so keep this one if you are using Windows.
 - app-mac.jar is for Mac with Intel CPUs. [How to check your Mac CPU type](#)
 - app-mac_aarch64.jar is for the newer Apple Silicon M1/M2 chips. [How to check your Mac CPU type](#). [List of Mac models with Apple Silicon CPUs](#)
 - app-linux.jar should work with any linux distro
6. Click on Main and use the run button to run your project
7. Run Poland first, as it's already done for you. Look at the code in Lab03Flagmaker.java to see how it works.
8. Do each of the flags, test them by running Main and clicking them from the Flags menu (test several sizes), and then submit your assignment.
9. To submit, zip or compress your entire jspacco-lab03 folder and upload the zipfile to Google Classroom.

Advice

- **Draw a diagram!** If you don't need a diagram, then you don't need help from the instructor or TAs. We've been programming longer than you have, and we all draw diagrams.
- You don't have to finish all of them to pass the assignment! See grading criteria at the end of the document.
- Note that your program should work for **any** flag of an allowable size, not simply the two example sizes given. So if the Colombian flag works for size 8, it should also work for size 12 and size 20.
- If you are struggling to come up with a solution to draw one of the flags, describe in words how you would break down the problem.
- Did you rename your project folder to username-lab03?

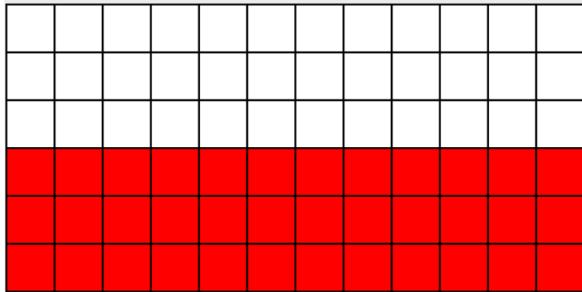
Flags

1. Poland

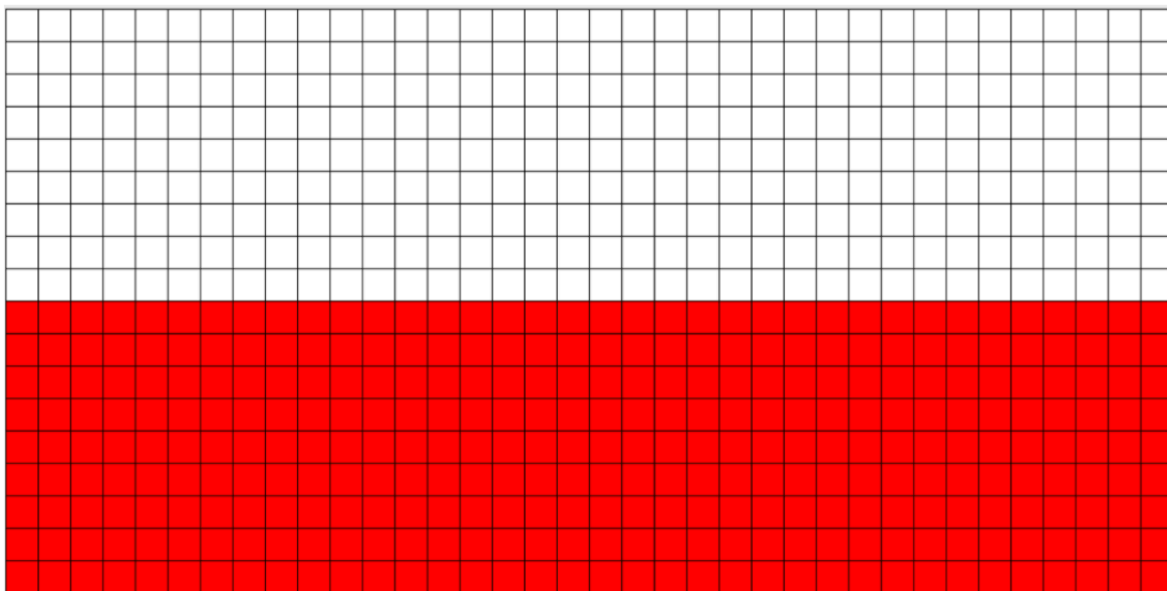


The flag of Poland is red and white, with the top half white and the bottom half red. The height should be even.

Size 6



Size 18



```
int w = grid.getNumCols();  
int h = grid.getNumRows();  
for (int c=0; c < w; c++) {
```

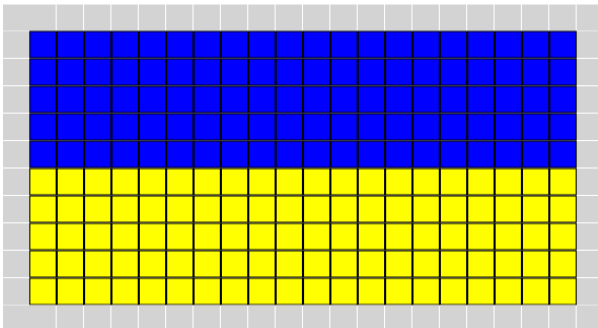
```
    for (int r=h/2; r < h; r++) {  
        grid.setColor(r, c, Color.RED);  
    }  
}
```

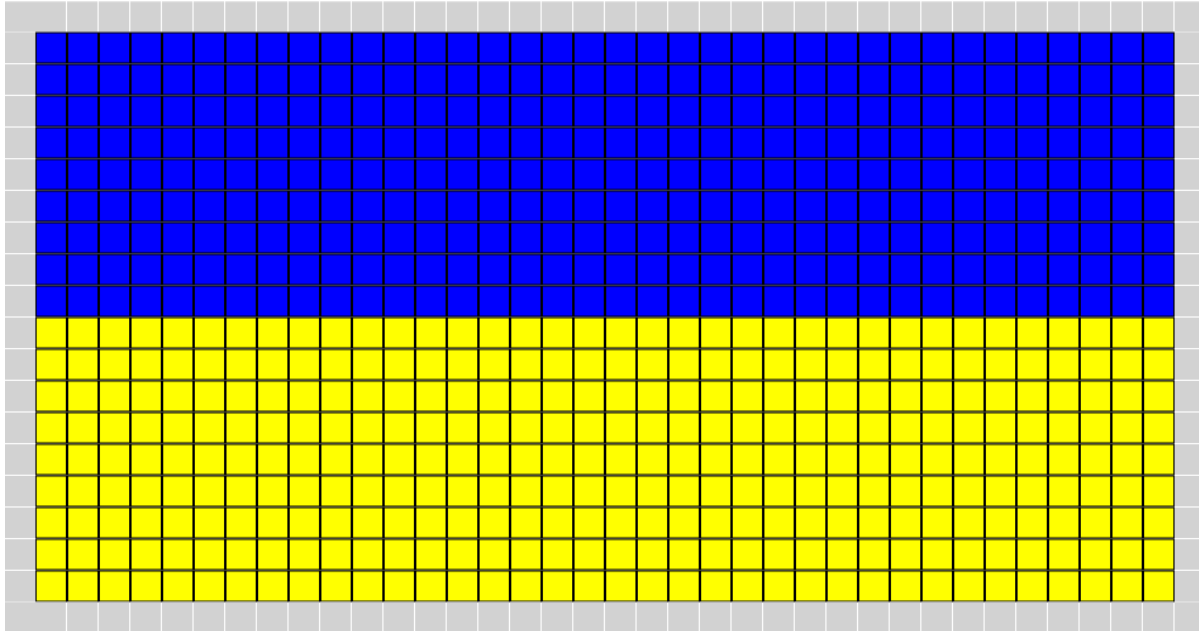
2. Ukraine (Slava Ukraini)



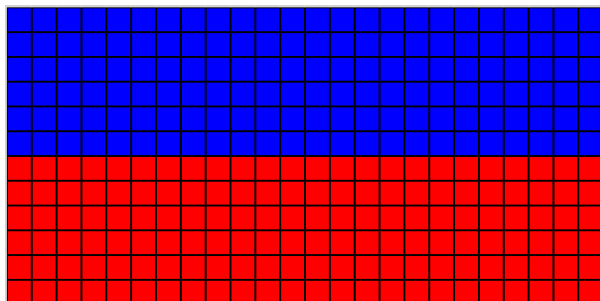
The flag of Ukraine is a horizontal bicolor of blue and yellow. The height of the flag must be even.

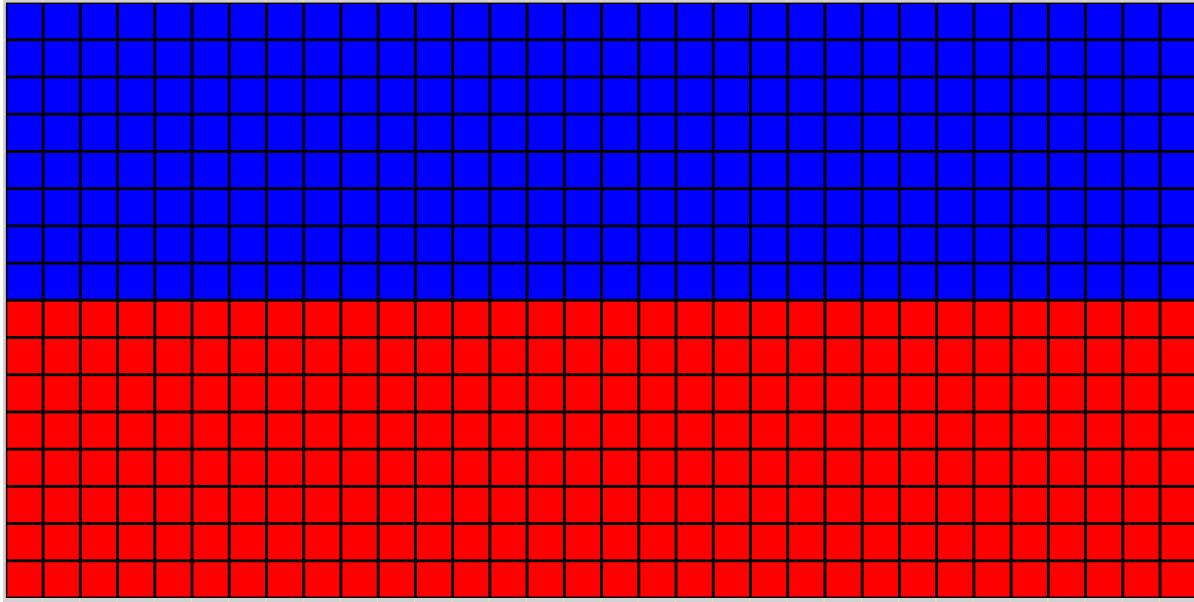
Size 10



Size 18**3. Haiti**

The flag of Haiti is a horizontal bicolor of blue and red. The height of the flag must be even.

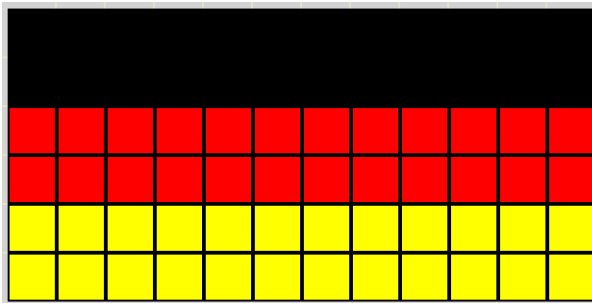
Size 12

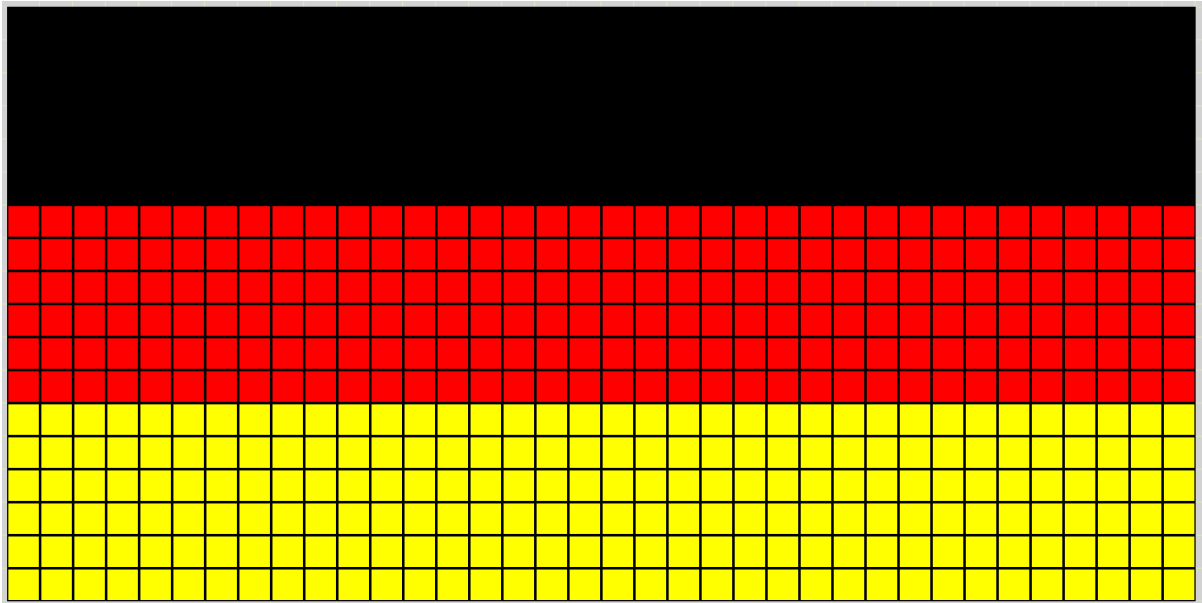
Size 16

4. Germany



The Germany flag is 3 equal sized horizontal stripes of black, red, and yellow. Any size that is a multiple of 3 should work. For example:

**Size 6**



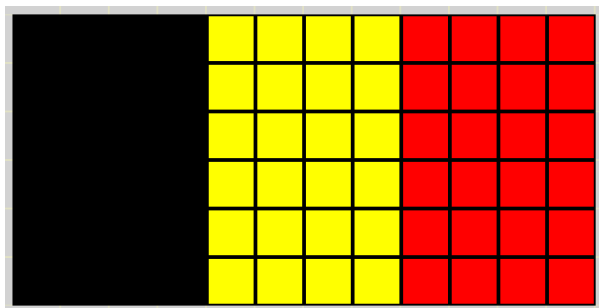
Size 18

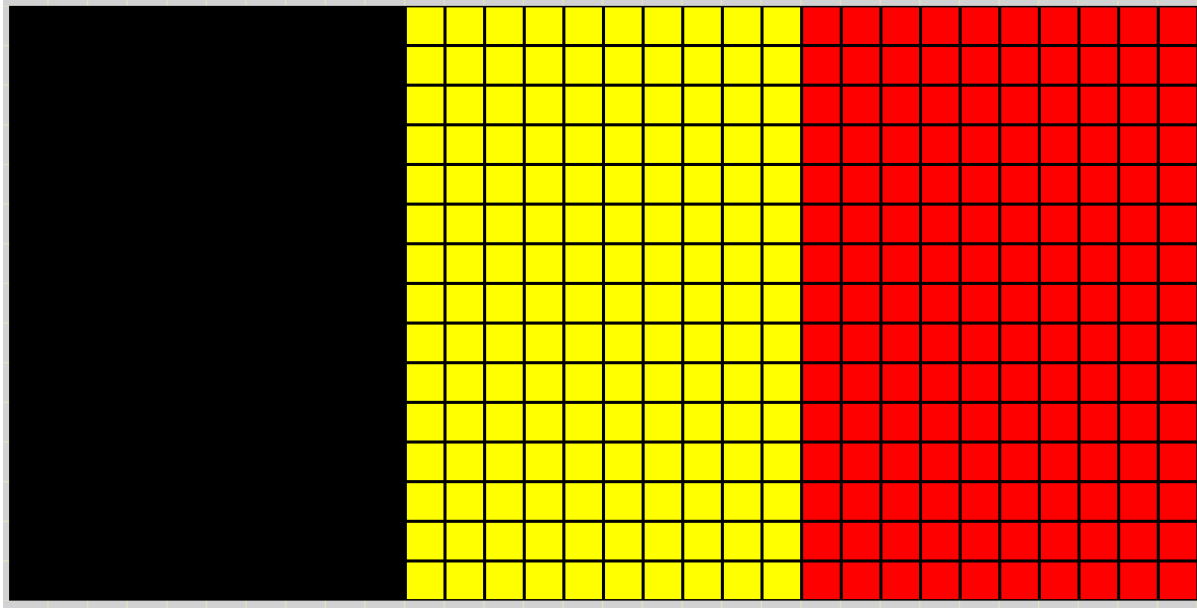
5. Belgium



The flag of Belgium is black, yellow, and red, with the black on the left, the yellow in the middle, and the red on the right. The height should be a multiple of 3.

Size 6

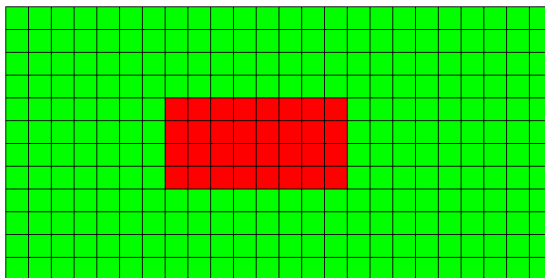


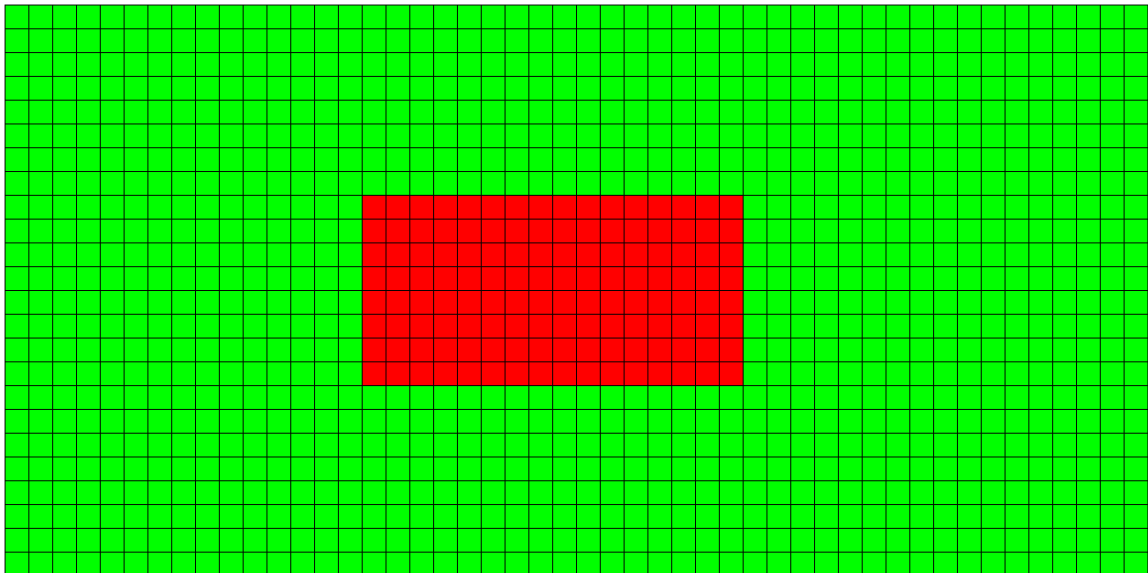
Size 15

6. Bangladesh



The flag of Bangladesh has a green background with a red circle in the middle (it's not technically in the middle but we are going to draw it in the middle). Drawing circles is very difficult, so we will approximate the circle with a rectangle. Your rectangular red dot should be in the middle third of the flag. The height and width of the flag must be a multiple of 3.

**Size 12**

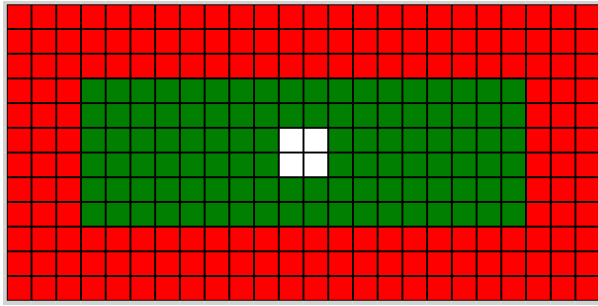
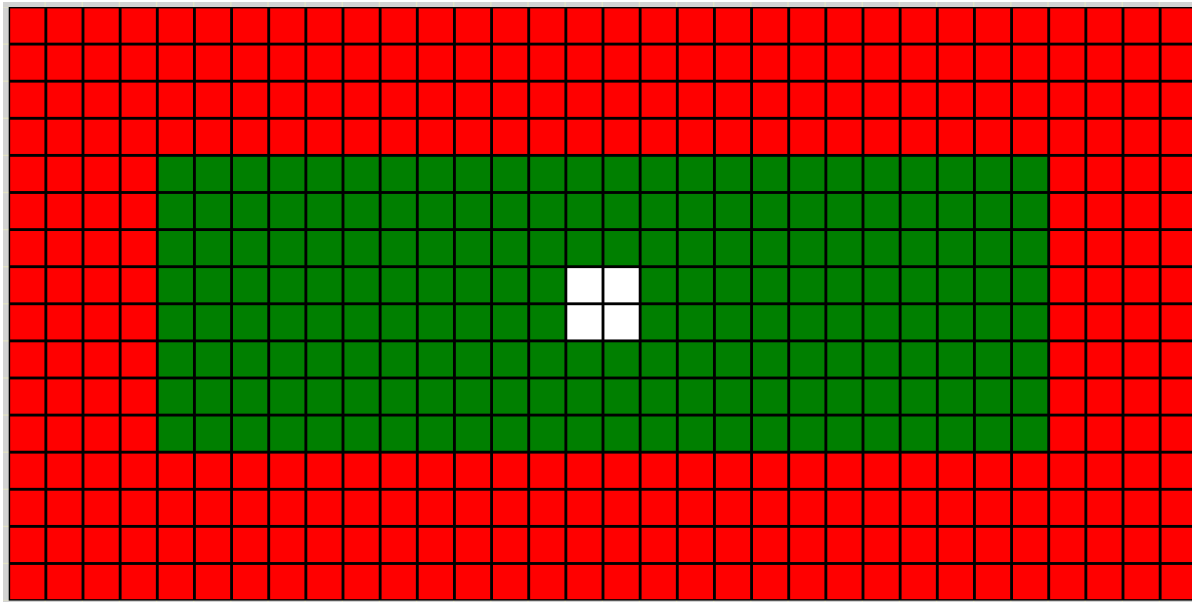


Size 24

7. Maldives

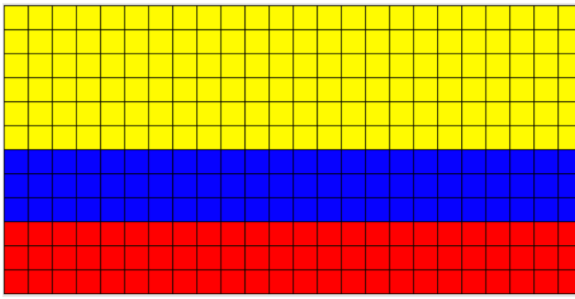


The flag of Maldives has a green flag field with a thick red border. In the center of the flag, there is a white crescent. The height should be a multiple of 4. We are going to approximate the crescent with a 2x2 white rectangle in the middle. We will also draw vertical red border 1/4 of height square thick and horizontal red border with 1/8 width (same as the 1/4 of height since grid width = 2 * grid height) of around the flag.

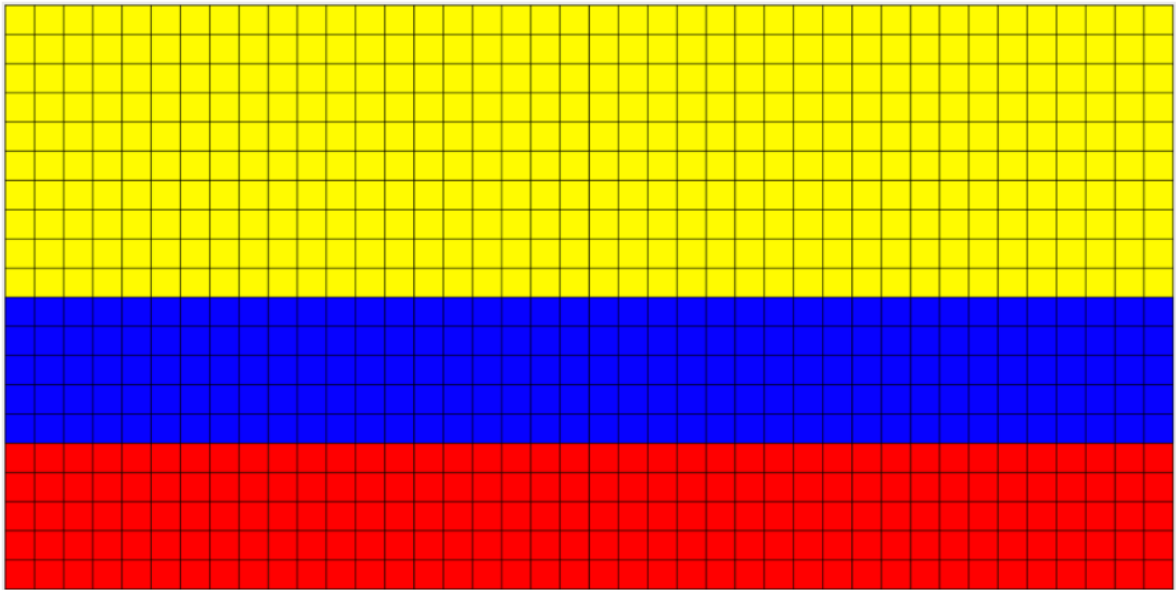
Size 12**Size 16****8. Colombia**

The Chilean flag is a horizontal bicolor of white and red. The height of the flag must be even. There is a blue square in the top left corner with a white star in the middle. We aren't going to draw the star, but we will draw the square. The square is as wide as it is tall.

Size 12



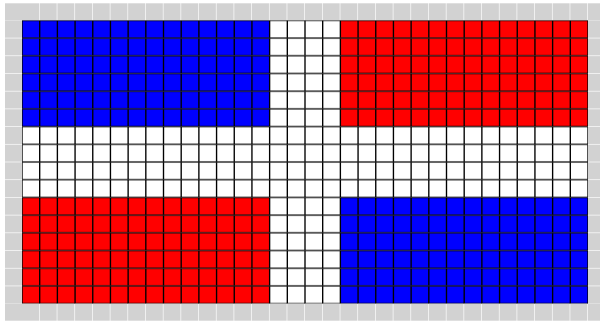
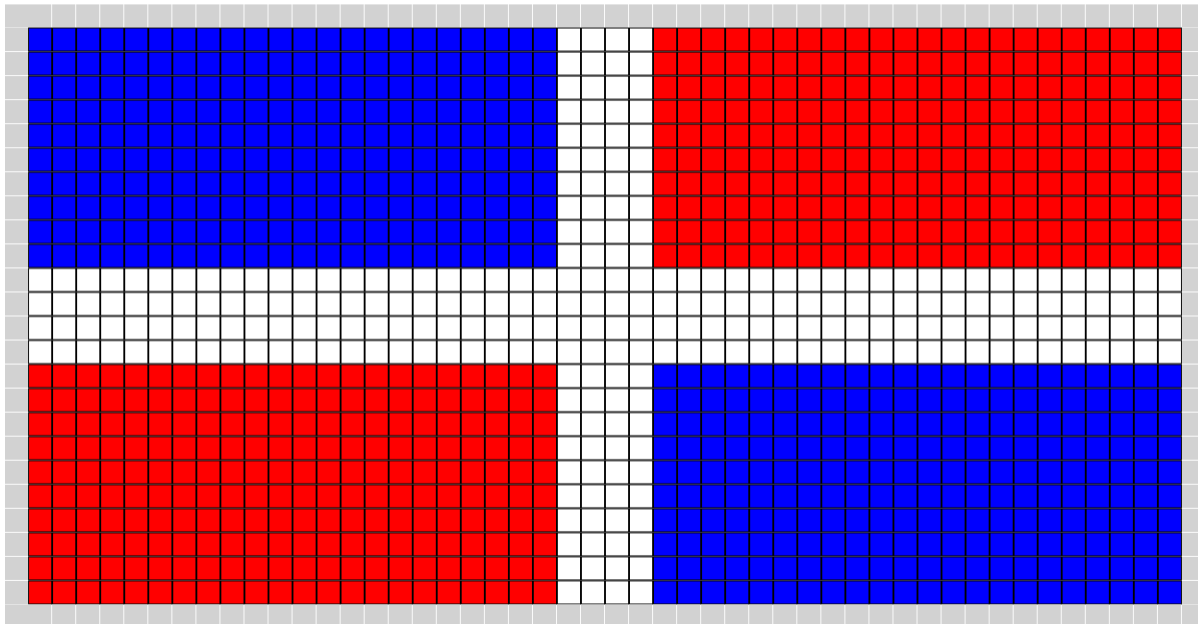
Size 20



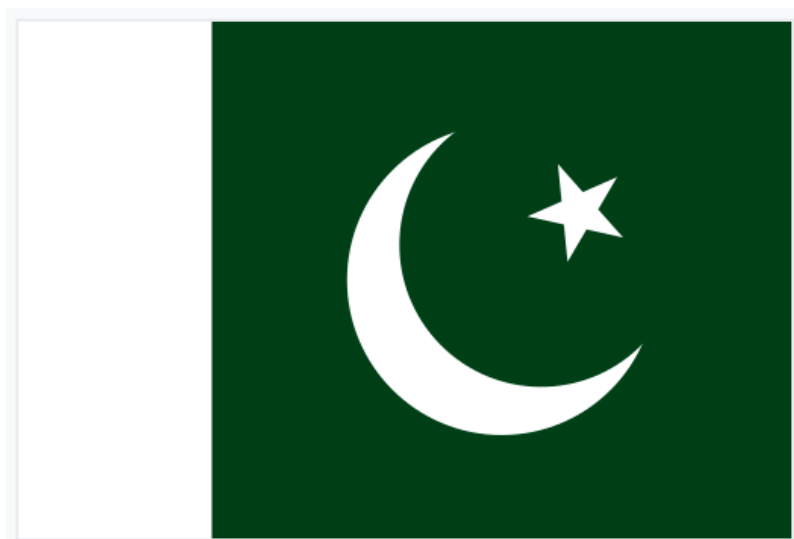
9. Dominican Republic



The Dominican Republic flag has blue in the top left and bottom right, and red in the top right and bottom left. There is a cross in the center. We are going to draw the white cross with a width of 4 no matter the size of the flag. The height of the flag should always be a multiple of 2, and at least 16 to look good. There is also a shield in the middle of the flag, but we are going to skip that part.

Size 16**Size 24**

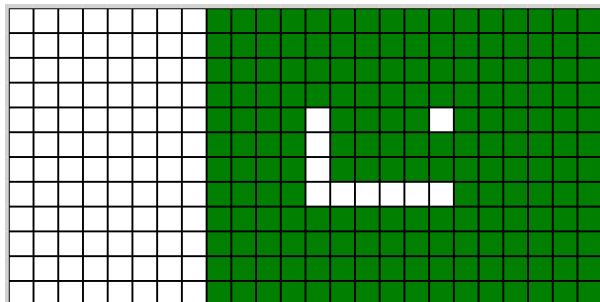
10. Pakistan



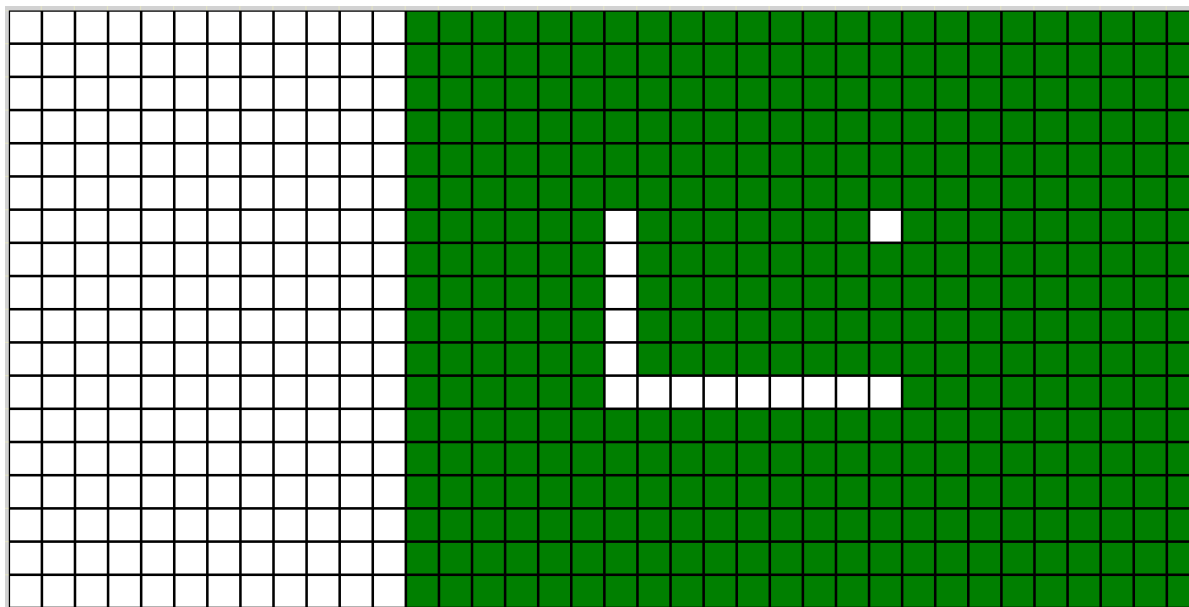
The flag of Pakistan is white and green, with the white on the left, the green on the right. On the green portion there is a crescent with a star. The height should be a multiple of 6. We are going to approximate the crescent with a white 'L' shape in the

middle followed by a white pixel on the opposite corner of "L" shape. The vertical portion of this "L" shape will be on the middle 3rd portion of the height (from $\frac{1}{3}$ (including) to $\frac{2}{3}$ (not including) of height) Similarly the horizontal portion of the "L" shape from the $\frac{1}{2}$ (including) to $\frac{3}{4}$ (not including) of the width. and the star is the corner of that "L" shape.

Size 12

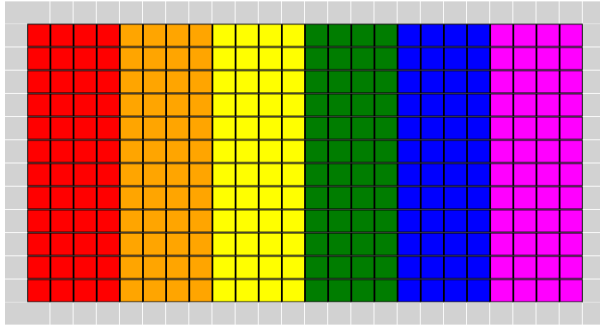
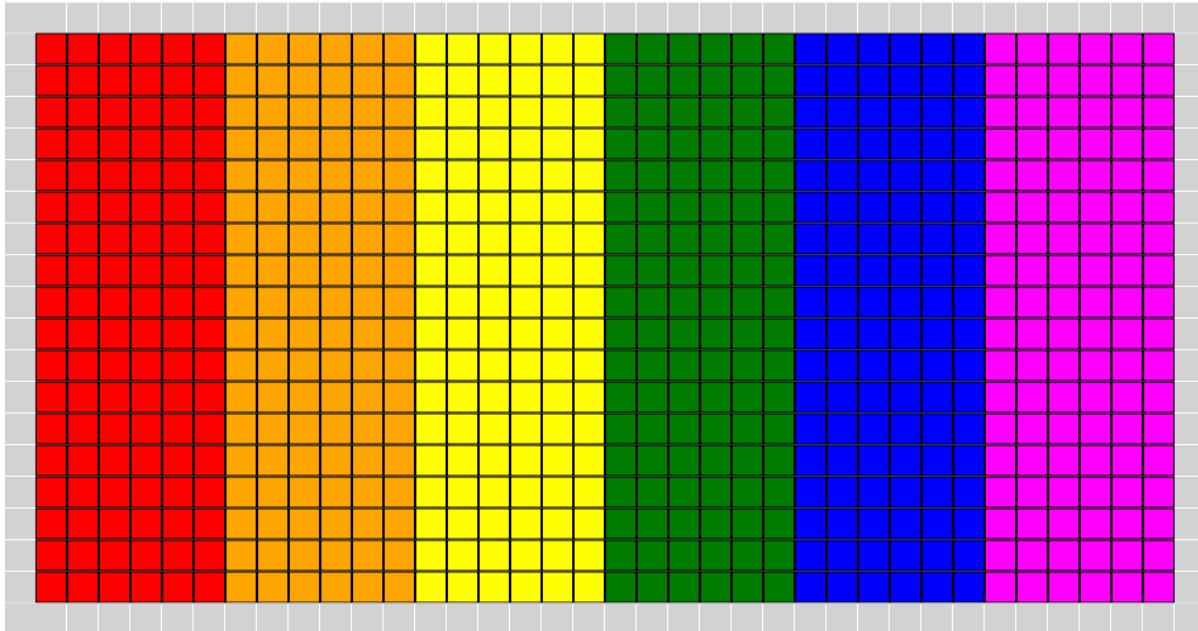


Size 18



11. Rainbow (rotated LGBTQ)

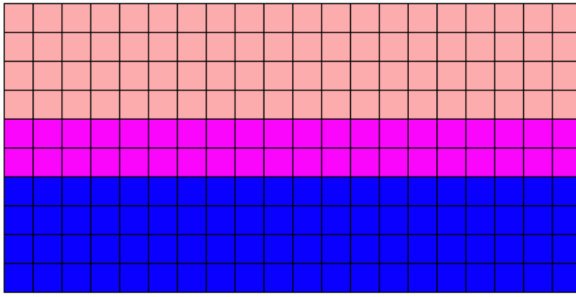
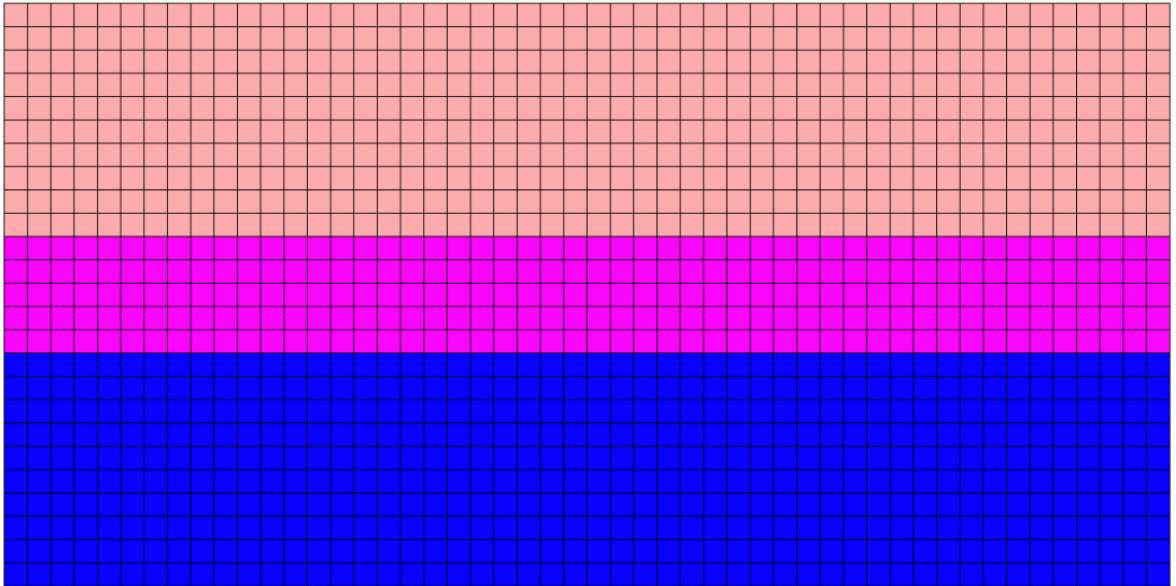
The LGBTQ pride flag has 6 colors: Red, Orange, Yellow, Green, Blue, and Magenta (purple). Normally the stripes are horizontal, but we are going to rotate them to make them vertical. Any multiple of 3 is a legal height (since the width will then be a multiple of 6)

Size 12**Size 18**

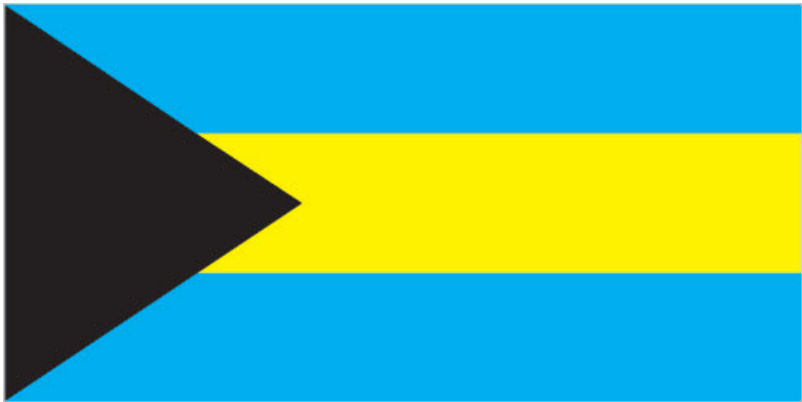
12. Bisexual



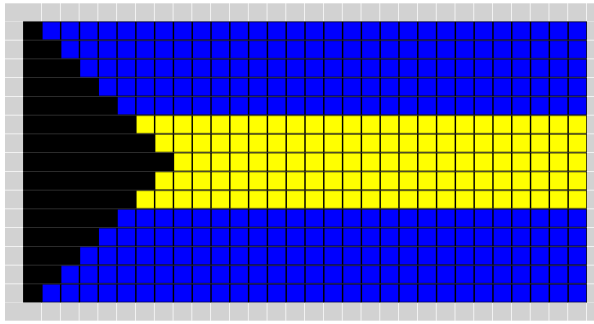
The Bisexual pride flag has pink at the top, blue at the bottom, and magenta (purple) in the middle. The ratio is 40% pink, 20% magenta, and 40% blue. This flag must be a multiple of 5 in height.

Size 10**Size 25**

13. Bahamas

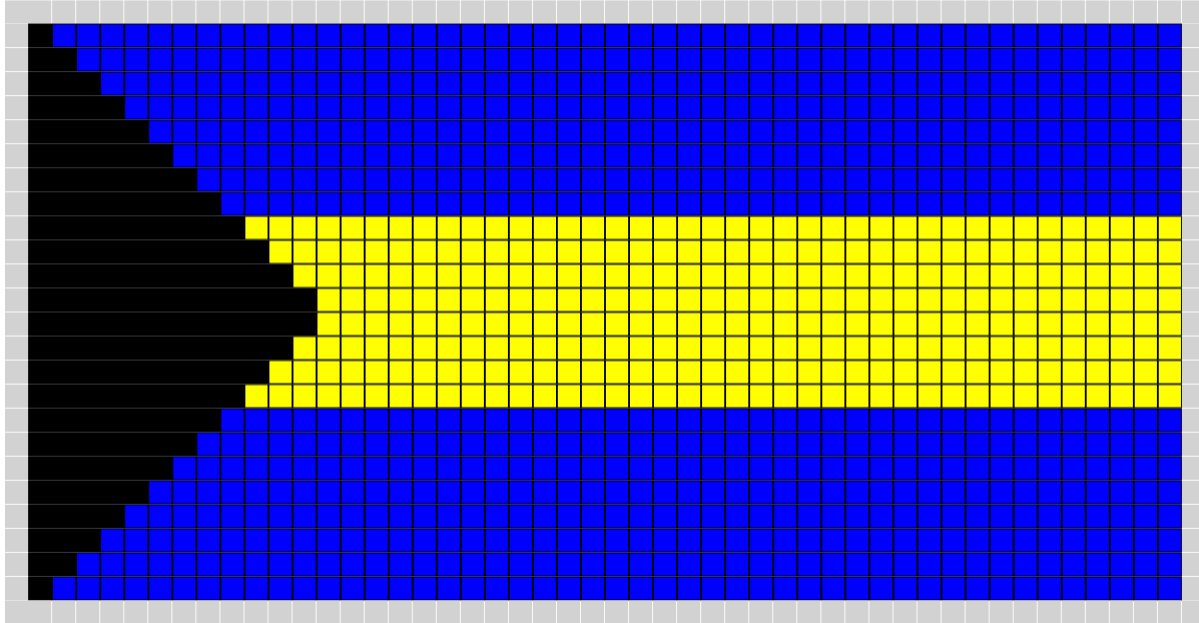


The Bahamas flag has a blue, yellow, and blue horizontal stripes, with a black triangle from the hoist (left) side. Any multiple of 3 should work.



TODO: Size 15

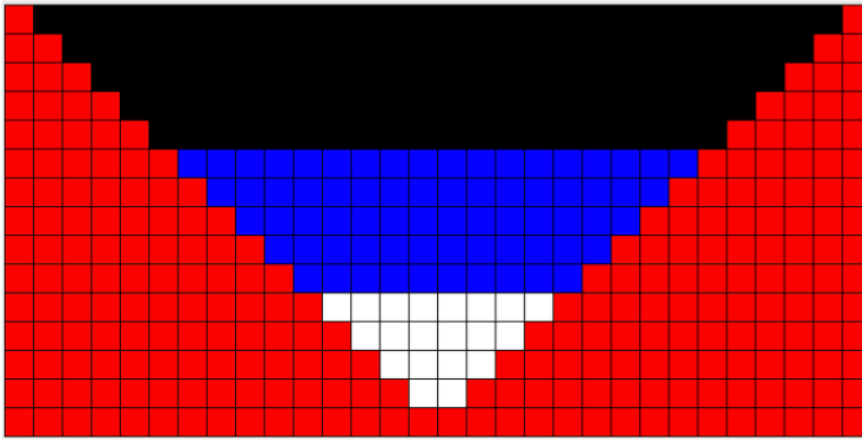
TODO: Size 24



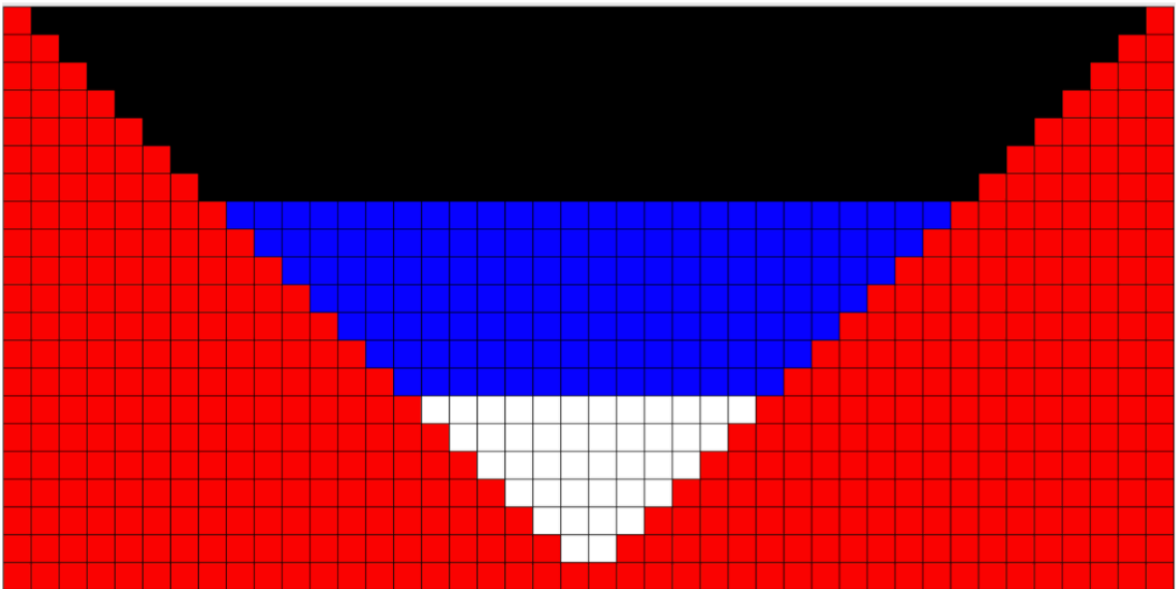
14. Antigua



Our approximation of the Antiguan flag has three stripes of black, blue, and white, each of the same width. There are two diagonal red shapes covering either side. The height must be a multiple of 3.



Size 15



Size 21

15. Macedonia



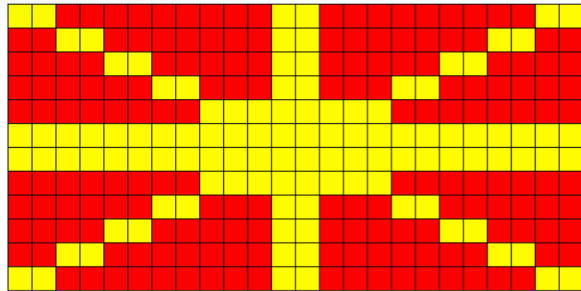
We will simplify the Macedonia flag in the following ways:

- We approximate the circle in the middle with a square that is 4 tall by 8 wide, regardless of how tall the flag is.

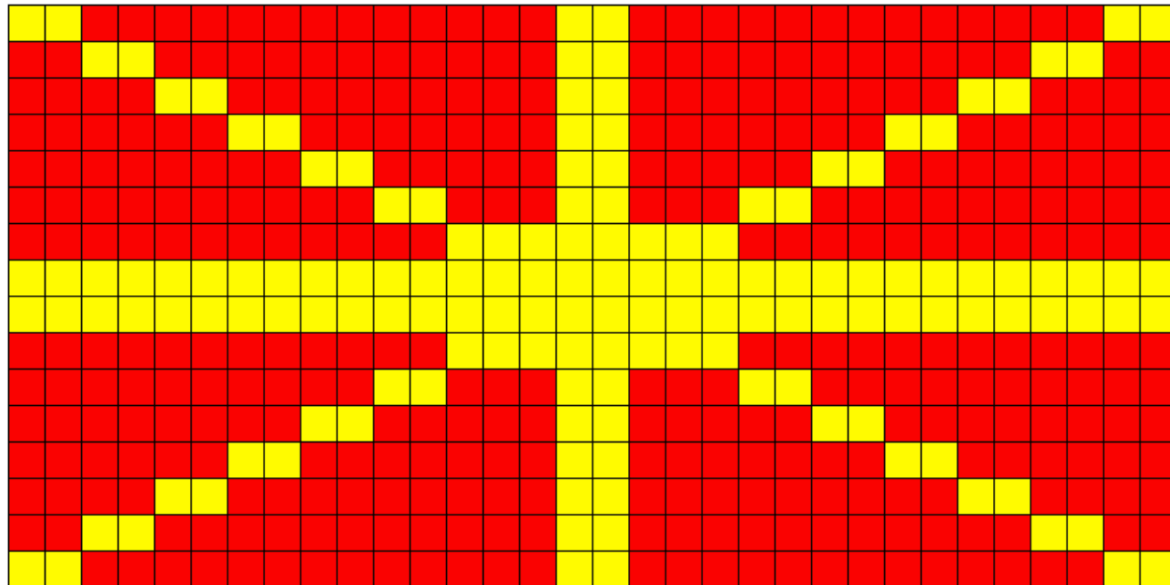
- We approximate the horizontal and vertical rays of sunshine with lines that are always 2 wide, regardless of the size of the flag.
- The flag is always a multiple of 4 in height.

Here are some examples:

Size 12



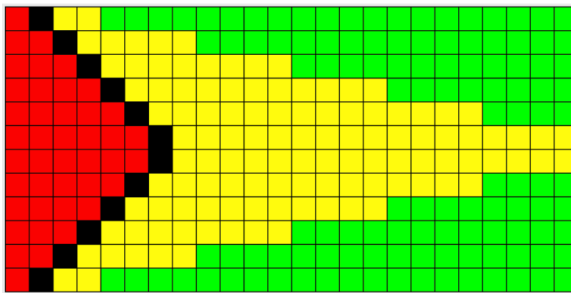
Size 16



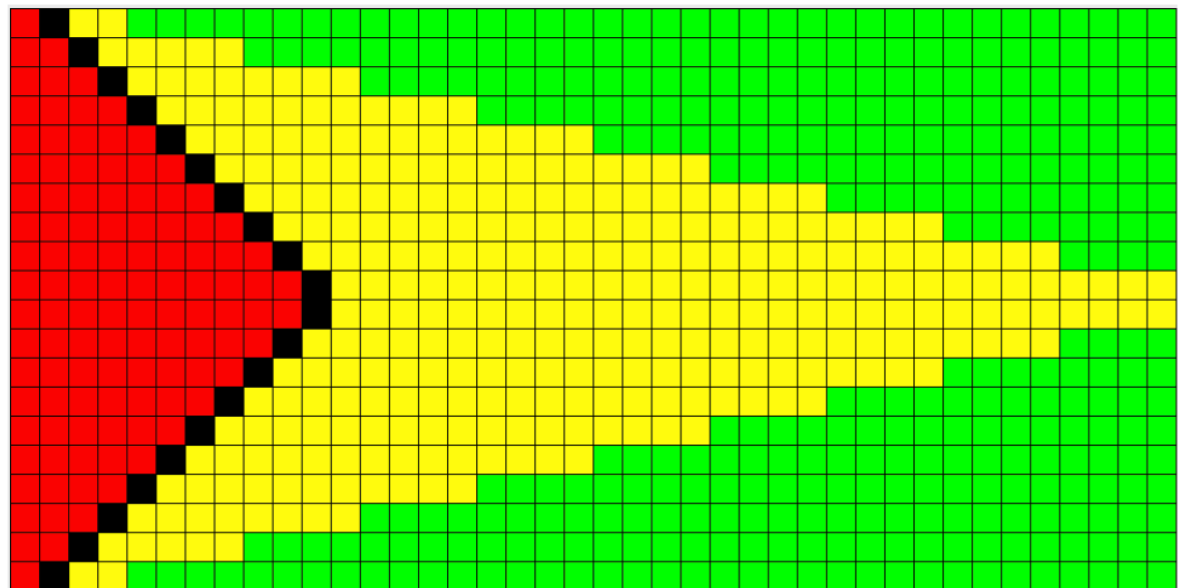
16. Guyana



The height of the flag of Guyana must be a multiple of four. Basically, you want triangles of yellow and red. Skip the white outline to the yellow segment of the flag.



Size 12



Size 20

Grading Rubric

- There are 15 total flags. You don't need to do them all to pass!
- The first 10 flags are worth 70 points (7 points each)

- Poland is still worth 7 even though it's already done for you
- Each additional flag is worth 5 points (11 flags is 80, 12 flags is 85, and so on)

Partial credit on flags

For the 5-point flags

- 5/5 flag works for multiple sizes
- 3/5 flag almost works for each size, or works for one size but not all sizes
- 1/5 there's effort but it's not really close to finished

For the 7 point flags

- 7/7 works for multiple sizes
- 5/7 almost works
- 3/7 there's effort